



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to GSN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

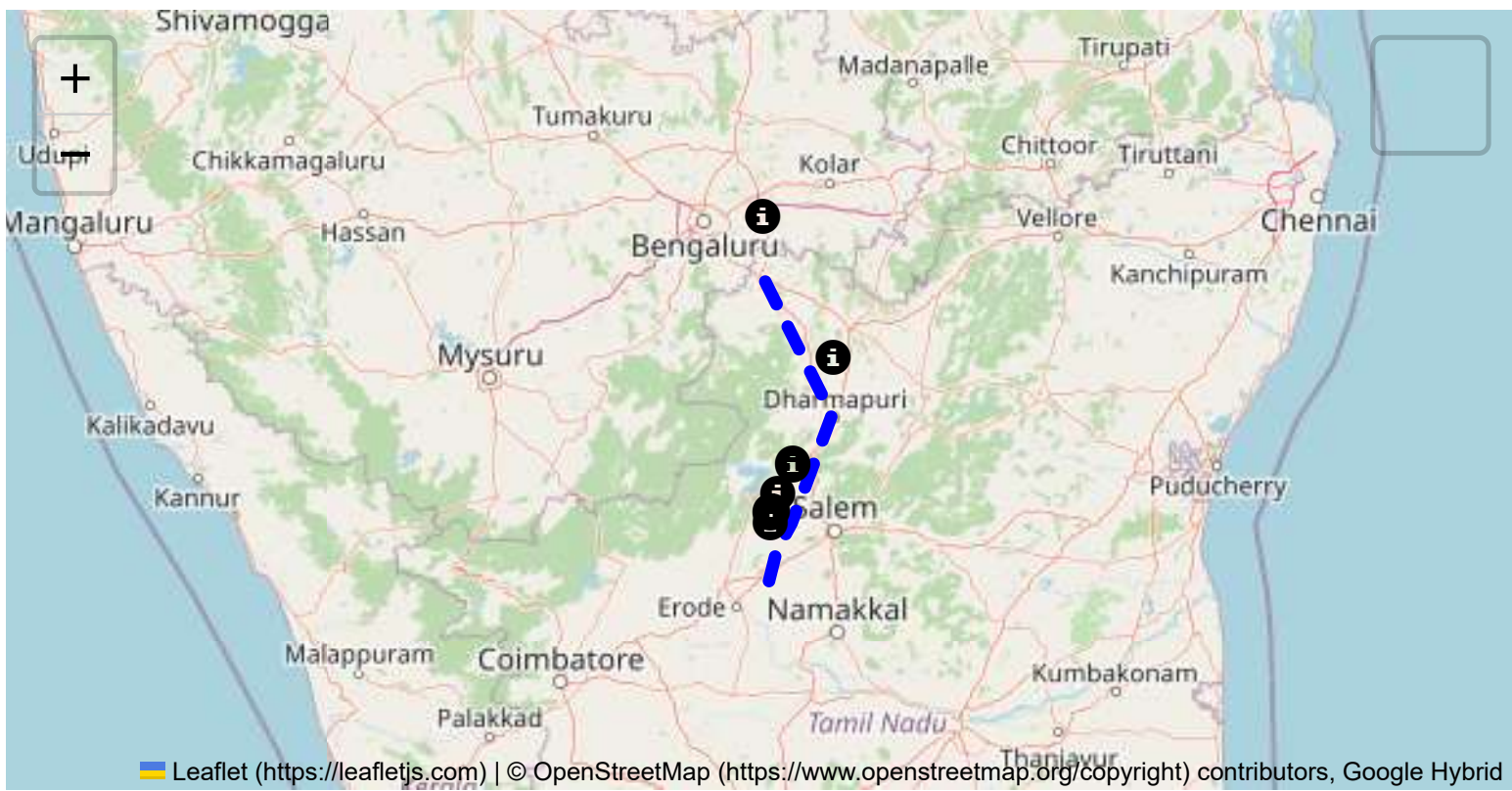
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

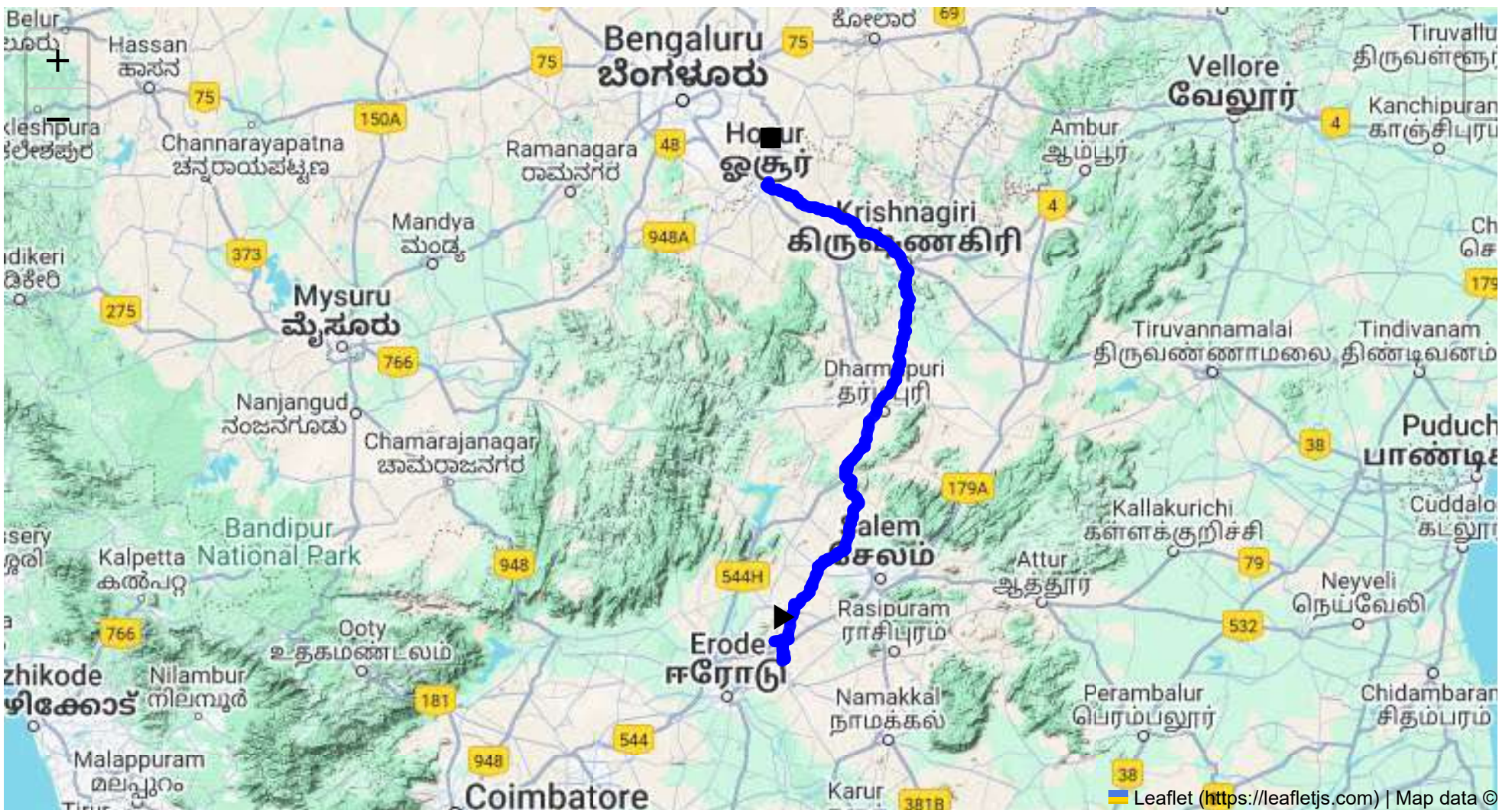
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 198.00 km
Estimated Duration: 3.9 hours
Adjusted Duration (Heavy Vehicle): 4.8 hours
Start: (11.4381, 77.8734)
End: (12.747125, 77.83356)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route begins at Sangagiri, passing through Edappadi, Konganapuram, Tharamangalam, and A.Reddihalli, and concludes in Hosur. This path primarily follows SH 86 and related roadways across approximately 198 kilometers, taking nearly 3.88 hours under typical conditions for heavy vehicles with hazardous materials.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The climate in Tamil Nadu is generally tropical, with hot summers and moderate winters. The monsoon season (June to September) brings heavy rains, which can cause flooding and poor visibility. This can impact driving conditions, particularly on rural roads, where waterlogging and landslides are possible.

3. Analysis of Traffic Patterns

Traffic tends to peak during morning (7:30 AM - 9:30 AM) and evening (5 PM - 7 PM) hours, especially near urban centers like Sangagiri and Hosur. Rural sections may see less traffic but can still have sporadic congestion near market areas, schools, or religious sites.

4. Assessment of Road Quality and Infrastructure

SH 86 and related roads generally have moderate to good quality with some potholes and uneven surfaces, particularly after the monsoon season. Narrow road sections may occur in rural areas, requiring cautious driving and limiting speeds for large vehicles.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, the National Highway 44 (NH 44) offers a more reliable, quicker route between Salem and Hosur for heavy vehicles. While slightly longer, NH 44 is better maintained and offers wider lanes.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu mandates specific permits for hazardous material transport, stipulating restrictions on timings and routes, especially in densely populated or eco-sensitive zones. It's essential to verify compliance with local authorities to avoid fines and ensure safety.

7. Overview of Historical Incidents

Although detailed incident-specific data is not readily available, there have been occasional reports of accidents involving heavy vehicles, primarily caused by driver error or brake failures. Regular checks and adherence to safety protocols are recommended.

8. Environmental Considerations and Sensitive Areas

The route may pass near ecologically sensitive regions such as agricultural lands and water bodies, necessitating precautions to prevent contamination. Noise and air pollution restrictions may also apply near residential zones.

9. Analysis of Communication Coverage

While communication infrastructure in Tamil Nadu is generally robust, rural areas might still experience dead zones. These should be anticipated in remote sections between major towns.

10. Estimated Emergency Response Times

Emergency services, particularly in rural stretches, are generally 30-45 minutes away. Urban areas have quicker response times due to better infrastructure, around 10-20 minutes.

11. Overall Summary of Risk Assessment

- **Route Condition:** Moderate with potential seasonal challenges.
- **Traffic & Regulations:** Manageable with adherence to local laws.
- **Communication & Emergency Response:** Some coverage gaps; response times vary significantly.
- **Risks:** Weather and environmental sensitivity are primary concerns.

It is recommended that drivers maintain heightened awareness during seasonal changes, ensure communications equipment is operational, and comply with local transport regulations diligently to mitigate risks. Trips should be planned to avoid peak traffic, optimize safe routing, and ensure environmental protection compliance.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	Medium	11.43995, 77.87431	30 KM/Hr
2	Turn	High	11.44107, 77.87528	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47932, 77.87006	30 KM/Hr
9	Turn	High	11.48267, 77.87250	15 KM/Hr
10	Turn	High	11.49278, 77.86821	15 KM/Hr
11	Turn	Medium	11.49338, 77.86642	30 KM/Hr
12	Blind Spot	Blind Spot	11.49000, 77.85120	10 KM/Hr
13	Turn	High	11.49430, 77.88504	15 KM/Hr
14	Turn	Medium	11.57477, 77.90536	30 KM/Hr
15	Turn	Medium	11.59143, 77.90623	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	11.59297, 77.90746	30 KM/Hr
17	Turn	Medium	11.63836, 77.94819	30 KM/Hr
18	Turn	High	11.64176, 77.95188	15 KM/Hr
19	Turn	Medium	11.64660, 77.95114	30 KM/Hr
20	Turn	High	11.69562, 77.97113	15 KM/Hr
21	Turn	Medium	11.71394, 77.99572	30 KM/Hr
22	Turn	Medium	11.71739, 78.00032	30 KM/Hr
23	Turn	Medium	11.71984, 78.01216	30 KM/Hr
24	Turn	Medium	11.72770, 78.02297	30 KM/Hr
25	Turn	High	11.73787, 78.03528	15 KM/Hr
26	Turn	High	11.73655, 78.03891	15 KM/Hr
27	Turn	High	11.74265, 78.04710	15 KM/Hr
28	Turn	Medium	11.78017, 78.05790	30 KM/Hr
29	Turn	Medium	11.86727, 78.08317	30 KM/Hr
30	Turn	Medium	11.91282, 78.05854	30 KM/Hr
31	Turn	Medium	11.92500, 78.06476	30 KM/Hr
32	Turn	Medium	11.93385, 78.06370	30 KM/Hr
33	Turn	Medium	11.93757, 78.05319	30 KM/Hr
34	Turn	Medium	11.95570, 78.04715	30 KM/Hr
35	Turn	High	11.95870, 78.04990	15 KM/Hr
36	Turn	High	11.96031, 78.04945	15 KM/Hr
37	Turn	High	11.96291, 78.05131	15 KM/Hr
38	Turn	Medium	11.96453, 78.05092	30 KM/Hr
39	Turn	High	11.96562, 78.04928	15 KM/Hr
40	Turn	Medium	12.17054, 78.16831	30 KM/Hr
41	Turn	Medium	12.23039, 78.19479	30 KM/Hr
42	Turn	Medium	12.24552, 78.19502	30 KM/Hr
43	Turn	Medium	12.25143, 78.20019	30 KM/Hr
44	Turn	Medium	12.25445, 78.20013	30 KM/Hr
45	Turn	Medium	12.29202, 78.19967	30 KM/Hr
46	Turn	Medium	12.30003, 78.20704	30 KM/Hr
47	Turn	Medium	12.32029, 78.20737	30 KM/Hr
48	Turn	Medium	12.39966, 78.21764	30 KM/Hr
49	Turn	Medium	12.40666, 78.22320	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
50	Turn	Medium	12.42571, 78.22661	30 KM/Hr
51	Turn	Medium	12.50321, 78.22352	30 KM/Hr
52	Turn	Medium	12.54424, 78.20127	30 KM/Hr
53	Turn	Medium	12.59726, 78.13621	30 KM/Hr
54	Turn	Medium	12.61871, 78.08325	30 KM/Hr
55	Turn	Medium	12.63686, 78.07076	30 KM/Hr
56	Turn	Medium	12.67136, 77.97420	30 KM/Hr
57	Turn	Medium	12.67723, 77.96625	30 KM/Hr
58	Turn	Medium	12.71153, 77.88852	30 KM/Hr
59	Turn	High	12.73486, 77.82986	15 KM/Hr
60	Turn	Medium	12.73651, 77.82982	30 KM/Hr
61	Turn	Medium	12.73956, 77.83270	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
7	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
8	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
9	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
10	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
11	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
12	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
13	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
14	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
15	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
16	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
17	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
19	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
20	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
21	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
22	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
24	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
25	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
26	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
27	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
28	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
29	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
30	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
31	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
35	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
37	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
42	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
43	clinic	Joe Hospital	12.5209601, 78.2150623	30 km/h	Medium
44	clinic	Ohm Prakash Clinic	12.5206321, 78.2159682	30 km/h	Medium
45	clinic	Gokulam Clinic & Scans	12.5207976, 78.2162023	30 km/h	Medium
46	clinic	Srinivasa Clinic	12.520975, 78.215755	30 km/h	Medium
47	clinic	Dr. Anbalagan Clinic	12.5208366, 78.2160593	30 km/h	Medium
48	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium
49	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
50	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
51	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
56	pharmacy	Jaisheka Medical	12.7116723, 77.890613	30 km/h	Medium
59	pharmacy	H2 Pharmacy	12.7301802, 77.8401037	30 km/h	Medium
60	clinic	Meera Nursing Home, Krishnagiri	12.7326185, 77.8292795	30 km/h	Medium
63	hospital	Amma Hospital	12.7393554, 77.8326981	30 km/h	Medium
64	pharmacy	துளசி பார்மசி	12.7457796, 77.8339248	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
23	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
32	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
38	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
39	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
40	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
41	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
52	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
53	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium
54	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium
55	school	New Little Flower Matriculation Higher Secondary School	12.6607923, 78.0169012	30 km/h	Medium
57	college	Adhiyamaan College of Engineering	12.7152133, 77.8678715	30 km/h	Medium
58	marketplace	Bathalapalli Vegetable Market	12.7260477, 77.8598177	30 km/h	Medium
61	school	Vijay Vidyalaya School	12.7397242, 77.8308427	30 km/h	Medium
62	school	Maharishi Vidhya Mandir Primary School	12.7397254, 77.8324166	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44107, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46765, 77.87037



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47932, 77.87006



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.48267, 77.87250



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49278, 77.86821



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49338, 77.86642



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49000, 77.85120



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49430, 77.88504



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57477, 77.90536



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59143, 77.90623



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59297, 77.90746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63836, 77.94819



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64176, 77.95188



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64660, 77.95114



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69562, 77.97113



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71739, 78.00032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71984, 78.01216



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72770, 78.02297



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73655, 78.03891



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74265, 78.04710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78017, 78.05790



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86727, 78.08317



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91282, 78.05854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92500, 78.06476



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93385, 78.06370



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93757, 78.05319



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95570, 78.04715



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95870, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96031, 78.04945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96291, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96453, 78.05092



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96562, 78.04928



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17054, 78.16831



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.23039, 78.19479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24552, 78.19502



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25143, 78.20019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29202, 78.19967



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.30003, 78.20704



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32029, 78.20737



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39966, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42571, 78.22661



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50321, 78.22352



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.54424, 78.20127



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.59726, 78.13621



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.61871, 78.08325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.63686, 78.07076



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.67136, 77.97420



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.67723, 77.96625



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71153, 77.88852



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.73486, 77.82986



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.73651, 77.82982



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.73956, 77.83270
